



BIOLOGY CH.18 — MICROORGANISM — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



Topics in this chapter:

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1. Microorganisms & Water Pollution

- E. coli Coliform bacteria ki presence water pollution (faecal contamination) ka indicator hai. [Q1080]
- Salmonella Typhi = typhoid cause karta hai. Vibrio Cholerae = cholera (contaminated water/food). Mycobacterium Tuberculosis = tuberculosis. [Q1080]

2. Viruses - Characteristics & Diseases

- HIV = Human Immunodeficiency Virus. AIDS = disease (Acquired Immunodeficiency Syndrome), virus nahi. [Q1081]
- Viral diseases: Dengue, AIDS, Influenza, Polio, Measles, Chickenpox, Covid-19. [Q1081, Q1090]
- Virus = submicroscopic infectious agent, sirf living cells ke andar replicate kar sakta hai. Living + non-living dono characteristics hain. DNA ya RNA strands covered with capsid. [Q1084]
- Virus inactive rehta hai jab tak kisi cell ko infect nahi karta - ye uska "living character" hai. Outside host cell reproduce nahi kar sakta. [Q1084]
- Anthrax = bacteria (Bacillus anthracis) se hota hai, viral nahi. Elephantiasis = filarial parasites se. [Q1081]

3. Protozoa - Amoeba & Paramecium

- Paramecium: definite shape, specific spot (oral groove) se food intake, cilia se food ko wahan tak move karta hai. Unicellular ciliate protozoan. [Q1082]
- Amoeba: unicellular, no special excretory organs. Nitrogenous waste simple diffusion se remove hota hai. Contractile vacuole waste collect karke plasma membrane par rupture karke release karti hai. [Q1083]
- Amoeba ka asexual reproduction = Binary fission (parent do identical daughter cells mein split hota hai). [Q1096]

- Euglena = eukaryotic unicellular flagellated microorganism. [Q1082]
- Plasmodium = protozoa, single-celled, divides only in host cell. Malaria cause karta hai. [Q1082]

4. Bacteria - Shapes & Extremophiles

- Bacteria shapes: Spherical (Cocci), Rod (Bacilli), Spiral (Spirilla), Comma (Vibrios), Corkscrew (Spirochaetes). [Q1086, Q1095]
- Bacteria = unicellular prokaryotes, almost everywhere found. [Q1086]
- Bacteria extreme habitats mein reh sakte hain: hot springs (>100°C/volcanoes), deep sea, thermal vents, Antarctica ice. Viruses extreme conditions mein nahi reh sakte. [Q1095]
- Clostridium Botulinum: refrigerator ke andar bhi grow kar sakta hai aur lethal botulinum toxins produce karta hai (low-oxygen conditions mein). Nerve functions block karta hai → respiratory/muscular paralysis. [Q1089]
- E. coli: environment, foods, intestines mein found. [Q1089]
- Salmonella: intestinal tract disease. [Q1089]

5. Fungi & Yeast

- Yeast = bread mein daala jata hai taaki wo rise ho sake. Yeast sugar consume karke CO₂ gas aur ethanol produce karta hai. CO₂ gluten ki wajah se dough ke andar trapped rehti hai → dough rise hota hai. [Q1085]
- Yeast reproduction = Budding (new organism parent ke small outgrowth/bud se develop hota hai). [Q1092]
- Rhizopus = spore formation se reproduce karta hai. [Q1094]
- Agaricus = mushroom genus, multicellular organism. All plants, animals, fungi species multicellular hote hain. [Q1097]

6. Algae & Cyanobacteria

- Blue-green algae (Cyanobacteria): prokaryotic, autotrophic, thylakoid membranes contain karte hain. First photosynthetic oxygen-liberating organisms. [Q1088]
- Mitochondria, Lysosomes, Golgi bodies = eukaryotic cell organelles hain, isliye cyanobacteria mein nahi hote. [Q1088]
- Algae: chlorophyll-bearing, thalloid, autotrophic, largely aquatic (freshwater + marine). Moist stones, soils, wood mein bhi milte hain. Some fungi ke saath lichen banate hain. [Q1093]

7. Reproduction in Microorganisms

- Budding: Yeast, Hydra. [Q1092]

- Binary fission: Amoeba, Planaria, Leishmania. [Q1096]
- Multiple fission: Plasmodium. Nucleus repeatedly divide karta hai, many daughter cells produce hoti hain. [Q1094]
- Spore formation: Rhizopus, Aspergillus, Penicillium. [Q1094, Q1096]
- Conjugation: Paramecium (sexual process, two individuals exchange genetic material). [Q1092]
- Vegetative propagation: Ginger, Potato - roots/shoots/leaves se without reproductive organs. [Q1096]

8. Disease-Causing Microbes - Quick Summary

- Viral diseases: Chickenpox, Covid-19, Influenza, Polio, Measles, Dengue, AIDS. [Q1090]
- Bacterial diseases: Tuberculosis, Pneumonia, Anthrax, Typhoid, Cholera. [Q1090]
- Fungal diseases: Mycosis. [Q1090]
- Worm/Parasite diseases: Ascariasis, Filariasis (Elephantiasis). [Q1090]
- Poliomyelitis (Polio): virus se hota hai - causes muscle weakness/paralysis. [Q1090]

9. Decomposers & Environment

- Bacteria (and fungi) = decomposers. Dead plants/animals ko humus mein convert karte hain. Complex organic matter ko CO₂, water, nutrients jaise inorganic substances mein break down karte hain. [Q1087]

10. Other Important Facts

- Planaria: flatworm (phylum Platyhelminthes), simple eyes called eyespots/ocelli - light detect kar sakti hain lekin image nahi bana sakti. No skeleton, no circulatory/respiratory organs. Flat ribbon-like body. [Q1091]
- Liver fluke aur Tapeworm bhi Platyhelminthes se belong karte hain. [Q1091]
- Ascaris: most common roundworm infection in humans. [Q1091]
- Multicellular organisms = plants, animals, fungi. Unicellular = Paramecium, Cyanobacteria, Mycoplasma. [Q1097]

11. ⚠ Traps / Key Differences

- AIDS = disease hai, HIV = virus hai. [Q1081]
- Anthrax = bacterial, Elephantiasis = parasitic - dono viral nahi hain. [Q1081]
- Virus inactive rehta hai jab tak cell infect nahi karta - ye uska living character hai. [Q1084]
- Amoeba = simple diffusion (excretion), Paramecium = cilia (food movement). [Q1082, Q1083]
- Cocci = spherical shape. [Q1086]

- Bacteria extreme habitats mein reh sakte hain, viruses nahi. [Q1095]
- Clostridium Botulinum = fridge mein bhi grow karta hai aur deadly toxins produce karta hai. [Q1089]